SFARI Desktop Metrics

(unnamed stream)

Coordinates	43.685784, -72.236713
COMID / HUC8	9327042 / 01080104
Drainage area	1.0005 km2
HR reach watershed area	1.0005 km2
Reach length	1000.0 ft
Watershed basis	HR reach watershed (STAF site engine v0.5.0)
StreamCat reach	Mink Brook (COMID 9327042), 444 ft downstream

Discipline	Function	Metric	Desktop method	Value / status	Data source
Hydrology	Catchment hydrology	Impervious surface area	Impervious and agricultural cover, NLCD 2021 over the HR reach watershed (STAF site engine); StreamCat by COMID when the engine has no value	6.0% impervious, 2.1% agricultural land (HR reach watershed)	STAF site engine v0.5.0 (HR reach watershed)
Hydrology	Catchment hydrology	Road density	Road density, TIGERweb roads over the HR reach watershed (STAF site engine); StreamCat rddens by COMID when the engine has no value	1.71 km/km2 road density (HR reach watershed)	STAF site engine v0.5.0 (HR reach watershed)
Hydrology	Catchment hydrology	Land use change	Impervious cover change, NLCD 2001 to 2021 over the HR reach watershed (STAF site engine); StreamCat 2001 to 2019 by COMID when the engine has no value	impervious 5.5% (2001) -> 6.0% (2021), delta +0.5 pts (HR reach watershed)	STAF site engine v0.5.0 (HR reach watershed)
Hydrology	Catchment hydrology	Impoundments	NID dams and their normal storage in the HR reach watershed (STAF site engine); StreamCat dam density by COMID when the engine has no value	0 NID dam(s) in the watershed, 0 acre-ft normal storage	STAF site engine v0.5.0 (HR reach watershed)
Hydrology	Surface water storage	Wetland Coverage	Woody and herbaceous wetland, NLCD 2021 over the HR reach watershed (STAF site engine); StreamCat by COMID when the engine has no value	13.2% wetland (HR reach watershed)	STAF site engine v0.5.0 (HR reach watershed)
Hydrology	Reach inflow	Concentrated flow inputs	Road-stream crossings on the NHDPlus HR network in the HR reach watershed (STAF site engine); StreamCat rdcrs by COMID when the engine has no value. Outfalls and ditches need local MS4 layers	1 road-stream crossing(s), 1.00 crossings/km2 (HR reach watershed)	STAF site engine v0.5.0 (HR reach watershed)
Hydrology	Reach inflow	Local runoff diversions	State water-rights and diversions GIS (local review)	*Local review required*	waterdata.usgs.gov
Hydrology	Streamflow regime	Flow permanence	USGS NWIS zero-flow days at the nearest comparable gage; StreamCat HYD integrity by COMID when no gage qualifies	0.0% zero-flow days at gage 01140817 (1078 d)	USGS NWIS 01140817 ELIZABETH MINE DISCHARGE NR S STRAFFORD, VT
Hydrology	Streamflow regime	Flow statistics	USGS NWIS daily flow, flow-duration percentiles at the nearest comparable gage	Q10 0.2 / Q50 0.1 / Q90 0.07 cfs (gage 01140817)	USGS NWIS 01140817

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Hydrology	Streamflow regime	Channel natural flow regime	Upstream NID normal storage over the HR reach watershed (STAF site engine) with the gage base-flow ratio; StreamCat dam storage by COMID when the engine has no value	baseflow ratio Q90/Q50 = 0.583 (gage 01140817), upstream normal storage 0 m3/km2 (HR reach watershed)	STAF site engine v0.5.0 (HR reach watershed)
Hydrology	Streamflow regime	Artificial structures and inputs	USACE NID dams within about a mile; levees from the National Levee Database by hand	0 dam(s) within ~1 mi (also check NLD levees)	USACE NID (dams) + National Levee Database
Hydraulics	Low flow and baseflow dynamics	Low flow velocity	Manning solver in the cross-section tool at the low-flow discharge	*Run cross-section tool*	
Hydraulics	Low flow and baseflow dynamics	Low flow depth	3DEP cross-section and the Manning solver in the cross-section tool at the low-flow discharge	*Run cross-section tool*	
Hydraulics	High flow dynamics	Overbank Flow Frequency	Bank height ratio, reach median of nine 3DEP cross-sections on the assessment reach (STAF site engine)	bank height ratio 1.68 (reach median, 3DEP sections)	STAF site engine v0.5.0 (HR reach watershed)
Hydraulics	High flow dynamics	Peak Flow Capacity (Morphological Check)	Bankfull width from the regional regression and the entrenchment ratio, reach median of nine 3DEP cross-sections (STAF site engine)	regional bankfull width 3.1 m, entrenchment ratio 1.54 (reach median, 3DEP sections)	STAF site engine v0.5.0 (HR reach watershed)
Hydraulics	High flow dynamics	Bed Mobilization Frequency	Shields comparison of bed shear and critical shear in the cross-section tool (enter D50)	*Run cross-section tool*	
Hydraulics	Floodplain connectivity	Lateral Floodplain Inundation	USFWS NWI wetland features near the reach, with the entrenchment ratio from the 3DEP reach cross-sections (STAF site engine)	4.9 ac of NWI wetland within 150 m of the reach (18.4% of the strip), 0.25 ac of it riverine · entrenchment ratio 1.54 (reach median, 3DEP sections)	USFWS National Wetlands Inventory (STAF data bundle) + STAF site engine reach cross-sections
Geomorphology	Channel and floodplain dynamics	Bank Migration and Meander	Multi-date NAIP or Google Earth imagery; USGS DSAS or RivMAP; EarthExplorer (local review)	*Local review required*	earthexplorer.usgs.gov
Geomorphology	Channel and floodplain dynamics	Channel pattern	Sinuosity of the assessment reach on the NHDPlus HR network (STAF site engine); compare the planform on TopoView and multi-date imagery	sinuosity 1.061	NHDPlus flowline geometry
Geomorphology	Sediment continuity	Transport Capacity	Channel slope (NHDPlus HR) with agricultural cover and SSURGO soil K over the HR reach watershed (STAF site engine); StreamCat by COMID when the engine has no value	channel slope 0.0100 m/m, watershed agriculture 2% (HR reach watershed), soil K 0.31 (HR reach watershed)	STAF site engine v0.5.0 (HR reach watershed)
Physicochemistry	Light and thermal regime	Riparian Canopy Cover	Forest in the 100 m riparian buffer, NLCD 2021 (STAF site engine); StreamCat riparian forest by COMID when the engine has no value	48.4% forest in the 100 m riparian buffer	STAF site engine v0.5.0 (HR reach watershed)
Physicochemistry	Carbon processing	Riparian Corridor Width and Quality	Natural vegetation in the 100 m riparian buffer, NLCD forest, shrub, grassland and wetland (STAF site engine); StreamCat by COMID when the engine has no value	73.1% natural vegetation in the 100 m riparian buffer (forest + shrub + grassland + wetland)	STAF site engine v0.5.0 (HR reach watershed)

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Physicochemistry	Nutrient cycling	Vegetated Riparian Corridor Width	Natural vegetation in the 100 m riparian buffer, NLCD forest, shrub, grassland and wetland (STAF site engine); StreamCat by COMID when the engine has no value	73.1% natural vegetation in the 100 m riparian buffer (forest + shrub + grassland + wetland)	STAF site engine v0.5.0 (HR reach watershed)
Biology	Community dynamics	Riparian communities	Natural vegetation in the 100 m riparian buffer, NLCD forest, shrub, grassland and wetland (STAF site engine); StreamCat by COMID when the engine has no value	73.1% natural vegetation in the 100 m riparian buffer (forest + shrub + grassland + wetland)	STAF site engine v0.5.0 (HR reach watershed)
Biology	Watershed connectivity	Upstream and Downstream Barriers	USACE NID dams within about a mile, with StreamCat CONN integrity by COMID; the National Aquatic Barrier Inventory by hand	0 dam/barrier(s) within ~1 mi · CONN integrity 0.97 (StreamCat, COMID 9327042)	USACE NID (+ National Aquatic Barrier Inventory) + StreamCat CONN
Biology	Watershed connectivity	Dewatered or Intermittent Segments	NHDPlus HR flow permanence code with StreamCat HYD integrity by COMID and the base-flow index over the HR reach watershed (STAF site engine)	NHD flow permanence: intermittent · HYD integrity 0.98 (StreamCat, COMID 9327042) · base-flow index 48% (HR reach watershed)	NHDPlus FCODE + StreamCat HYD + STAF site engine base-flow index

Available: the value was pulled and is printed beside its function on the field forms. Pending: the STAF site engine is still running. Unavailable: neither engine nor service answered. Local review required and Run cross-section tool: the assessor supplies it. Watershed basis and StreamCat reach above say whose watershed the values describe.